



Spaceborne Electron Accelerators

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 - University of Vermont

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Parallel Efforts

- Now
 - LANL – investigating C-band-based structures
 - SLAC – investigating X-band-based structures
- Soon
 - Frequency downselect
 - Joint effort on gun, beam dynamics, etc.
- This talk focuses on LANL effort to date. Errors are ours, don't blame Sami, Emilio or Jeff.

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Outline

- Why accelerators in space?
- Constraints
- Approach
- Results to date
- Future plans
- Conclusions

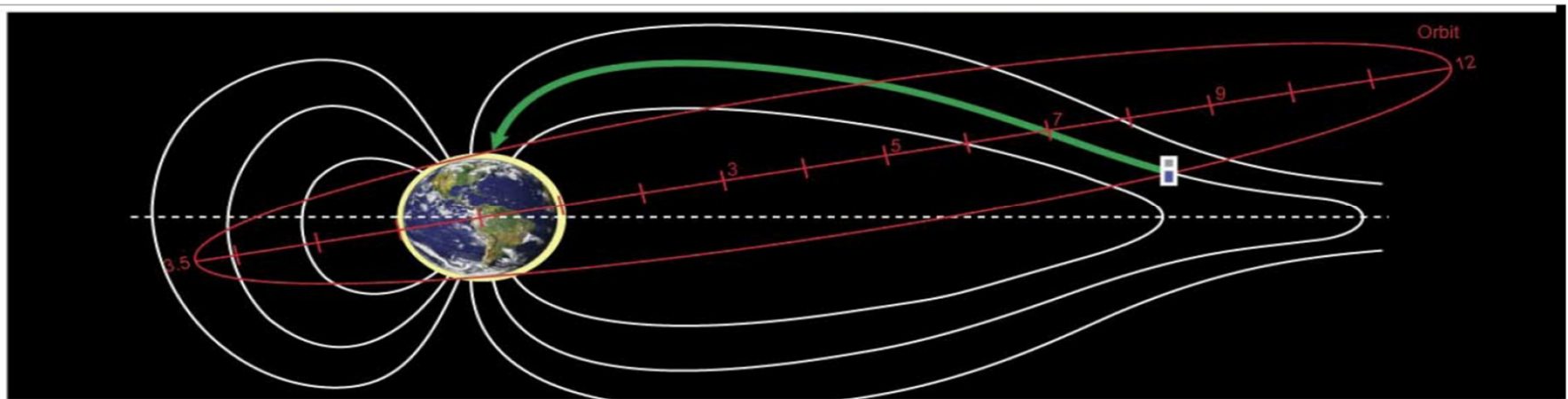
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Why Spaceborne Electron Accelerators?

Connections: How are the auroral ionosphere and nightside magnetosphere connected through the time-varying magnetic field?

We have magnetosphere *models*, but need better *measurements*.

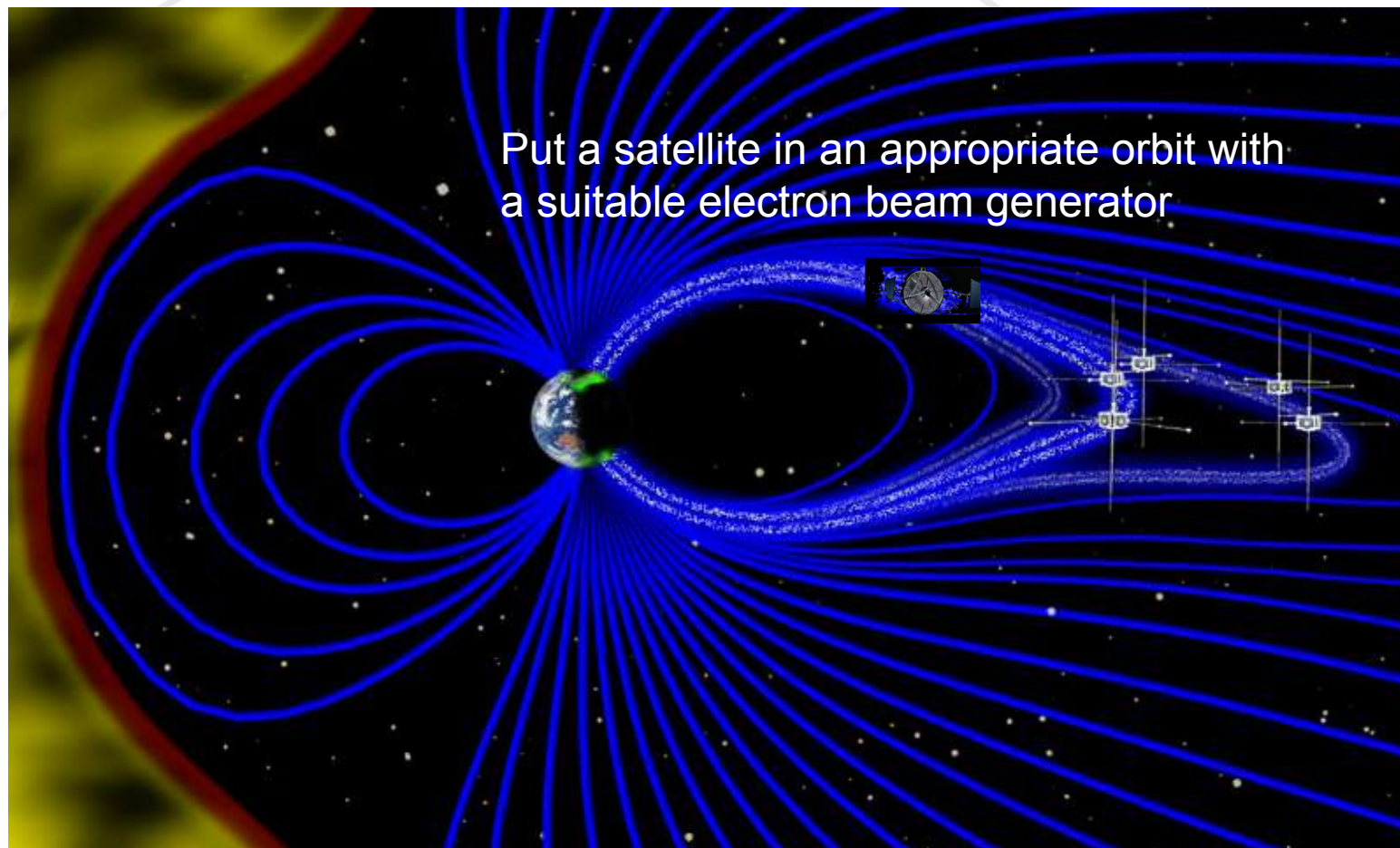


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Making the Measurement (1)



Emmanuel Masongsong/UCLA EPSS/NASA

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Making the Measurement (2)

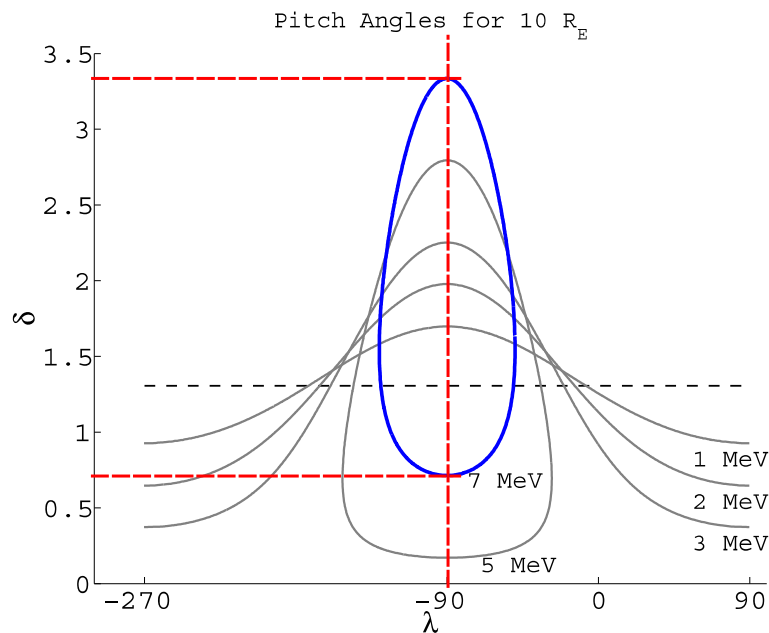


Image courtesy United Artists

Deliver ~10 kJ into the loss cone

More at:

“Modification of the loss cone for energetic particles”

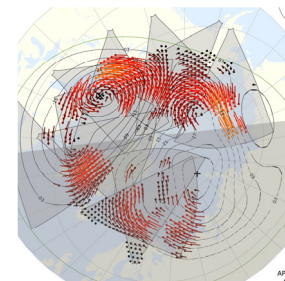
Porazik, Johnson, Kaganovich, Sanchez, GRL, 2014

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Making the Measurement (3)



Radar detection



Optical detection

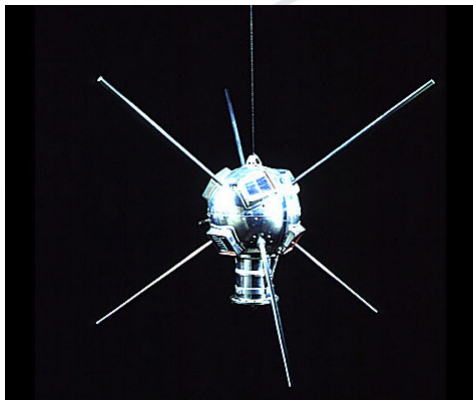
Image courtesy NASA

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Consider a Spherical Satellite...



$$\begin{aligned}
 10 \text{ kJ} &= 200 \text{ mA} \times 5 \text{ kV} \times 10 \text{ sec} \rightarrow 2 \text{ C} \\
 &20 \text{ mA} \times 50 \text{ kV} \times 10 \text{ sec} \rightarrow 0.2 \text{ C} \\
 &1 \text{ mA} \times 1000 \text{ kV} \times 10 \text{ sec} \rightarrow 0.01 \text{ C}
 \end{aligned}$$

For the same beam *power*, charge neutralization is easier with higher beam *voltage*.

But: DC voltages drive local plasma currents and corresponding currents in the satellite. Can lead to damage and breakdown.

For some satellites, 300 V = “high voltage”

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Other Constraints

- Minimize Size, Weight & Power (SWaP)
- Redundancy
 - as few “single points of failure” as possible

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So ... what to do?

- ~~High-voltage DC gun~~
- RF Linac: $P_{RF} = L \cdot \left(\frac{E_z^2}{R_s} + E_z I_b \right),$
 - Increase R_s as much as possible
 - Use modest gradients – lower RF power, longer linac to get to a given voltage
 - Moderate to heavy beam loading

Approach

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RF Linac: Bounding the Design

- Nominal 1-MV beam
- High power: 10 kW peak, 10% duty factor
- Efficient (for an RF linac)
 - 50% RF power to beam, 50% to structure
 - 50% RF generator efficiency
- **Linac draws 40 kW peak, 4 kW average**

Approach

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Can we Eliminate High Voltage?



Wolfspeed (formerly Cree) High Electron Mobility Transistors

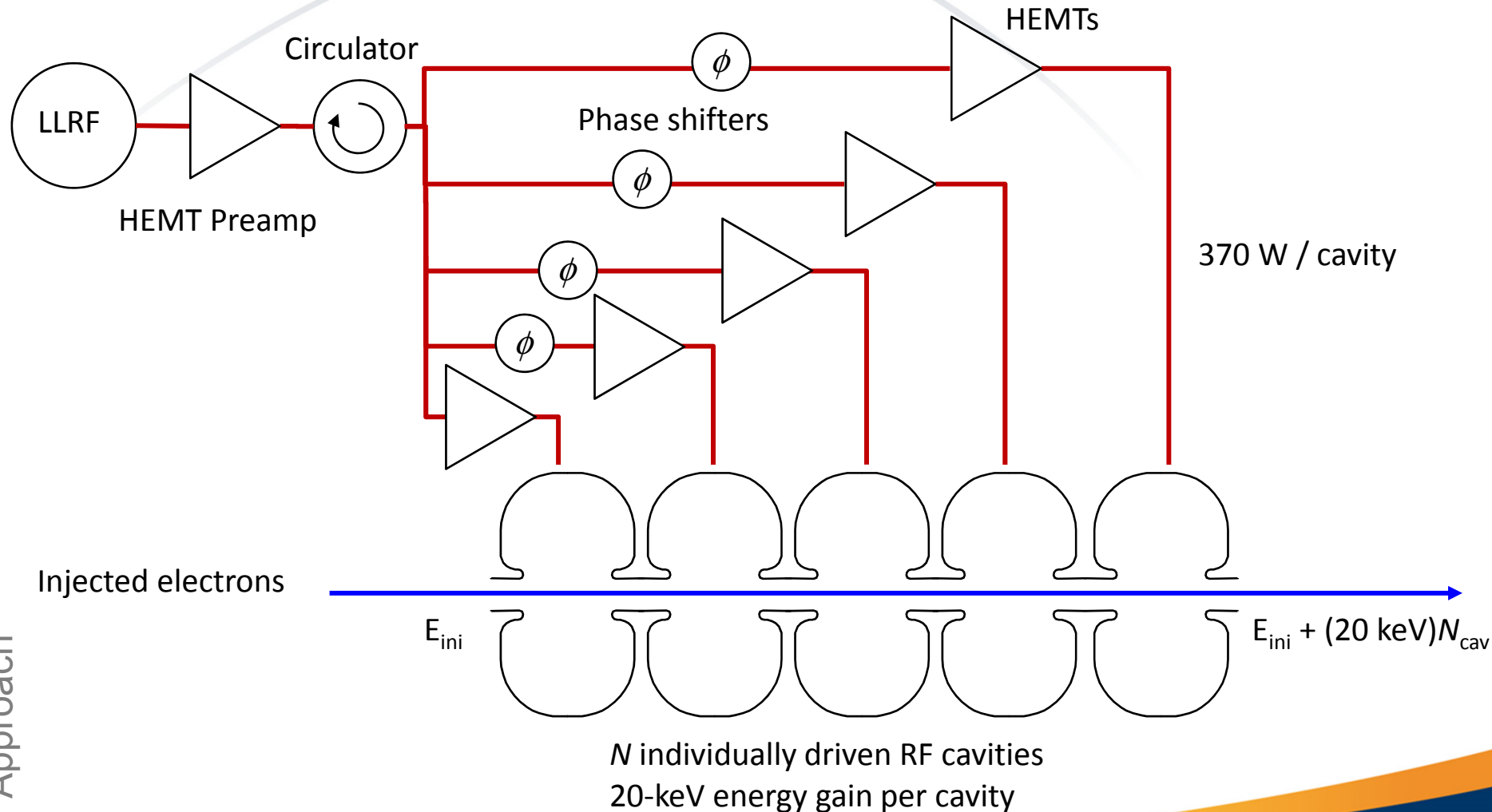
- up to 500 W @ 10% DF (C-band)
- power from 50 V DC
- up to 100 W @ 10% DF (X-band)
- >50% efficient

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Distributed vs. Monolithic Power

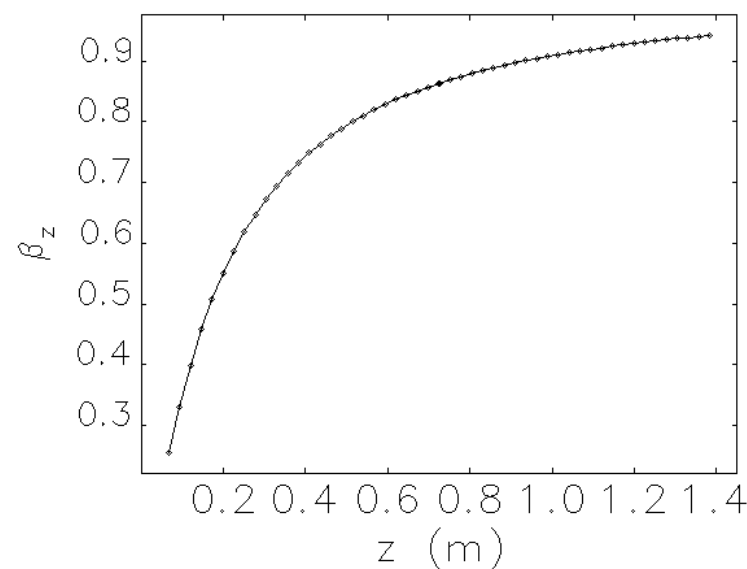
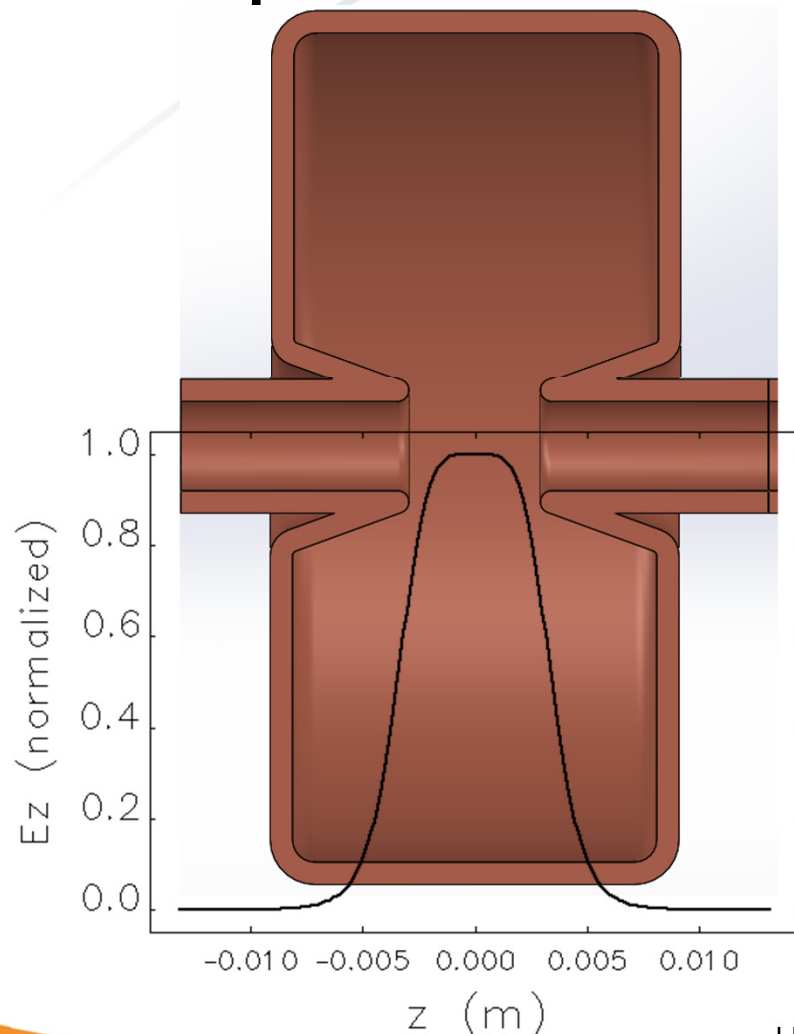


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Building Block: Low- β C-Band Cavity

- resonant at 5.1 GHz
- Power vs. energy gain: 170 W provides 10 – 20 kV gain per cell, depending on incoming beam voltage



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Thermal Loading: No Cooling System

- 50 cavities, 0.1 kg/copper cavity, 5 kg total structure mass
- Copper heat capacity, $C_{h,Cu} = 385 \text{ J/(kg K)}$
- Structure absorption = 4.25 J per macropulse
- There are 2,000 macropulses in each 10-s burst
 - 4.4 K/(10-s burst)
- Frequency detune $\sim -100 \text{ kHz/}^\circ\text{K @ 5.1 GHz}$
 - $\sim 44 \text{ kHz frequency droop per second}$
- **Total frequency detune in 10 seconds $\sim -440 \text{ kHz}$**
- **Cavity bandwidth $\sim 600 \text{ kHz}$**

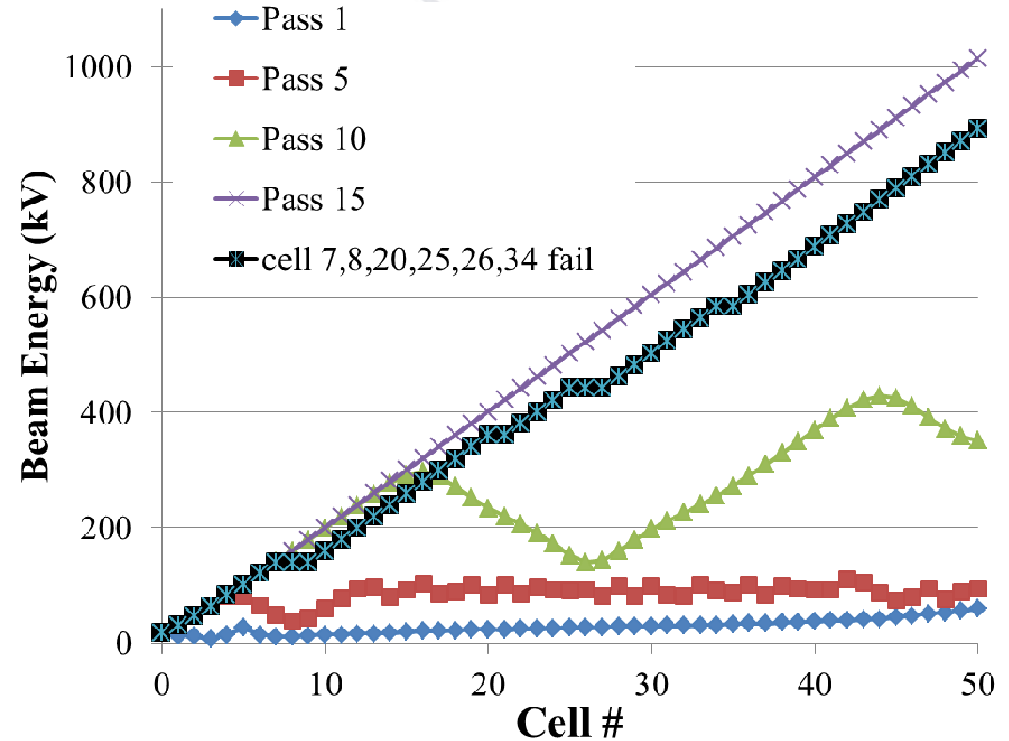
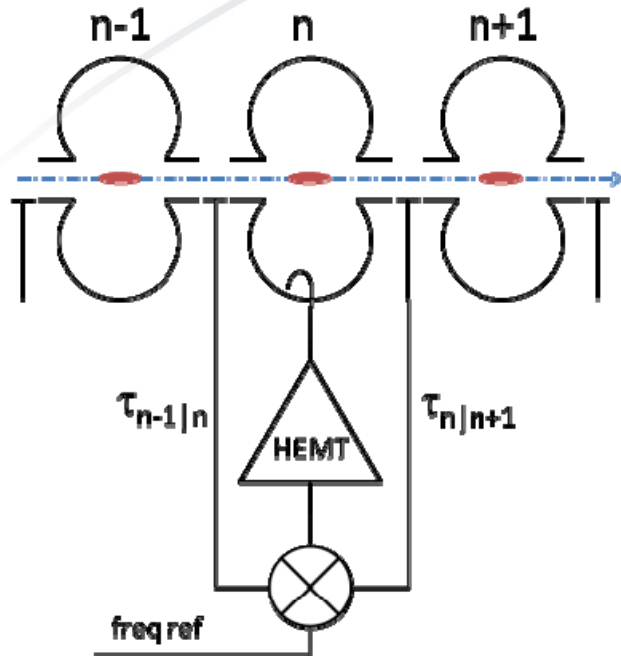
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Automatic Phase Control



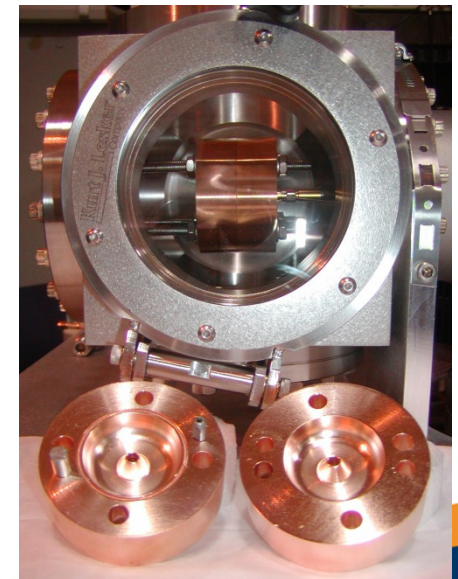
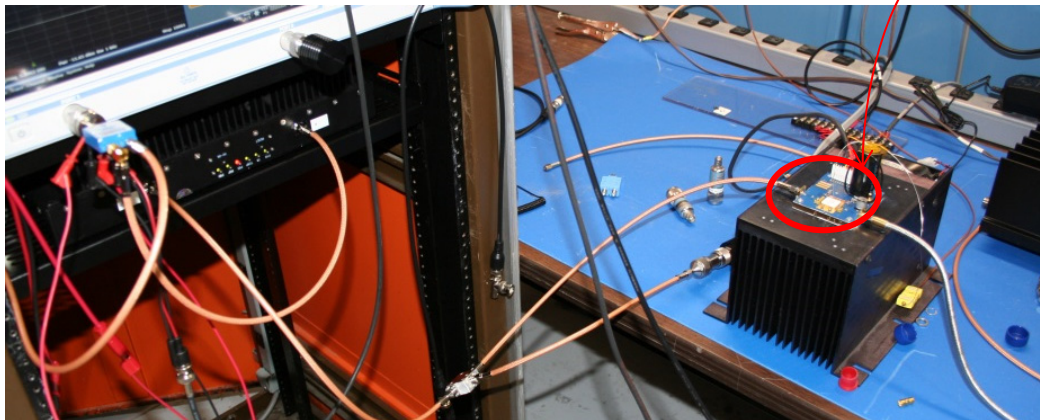
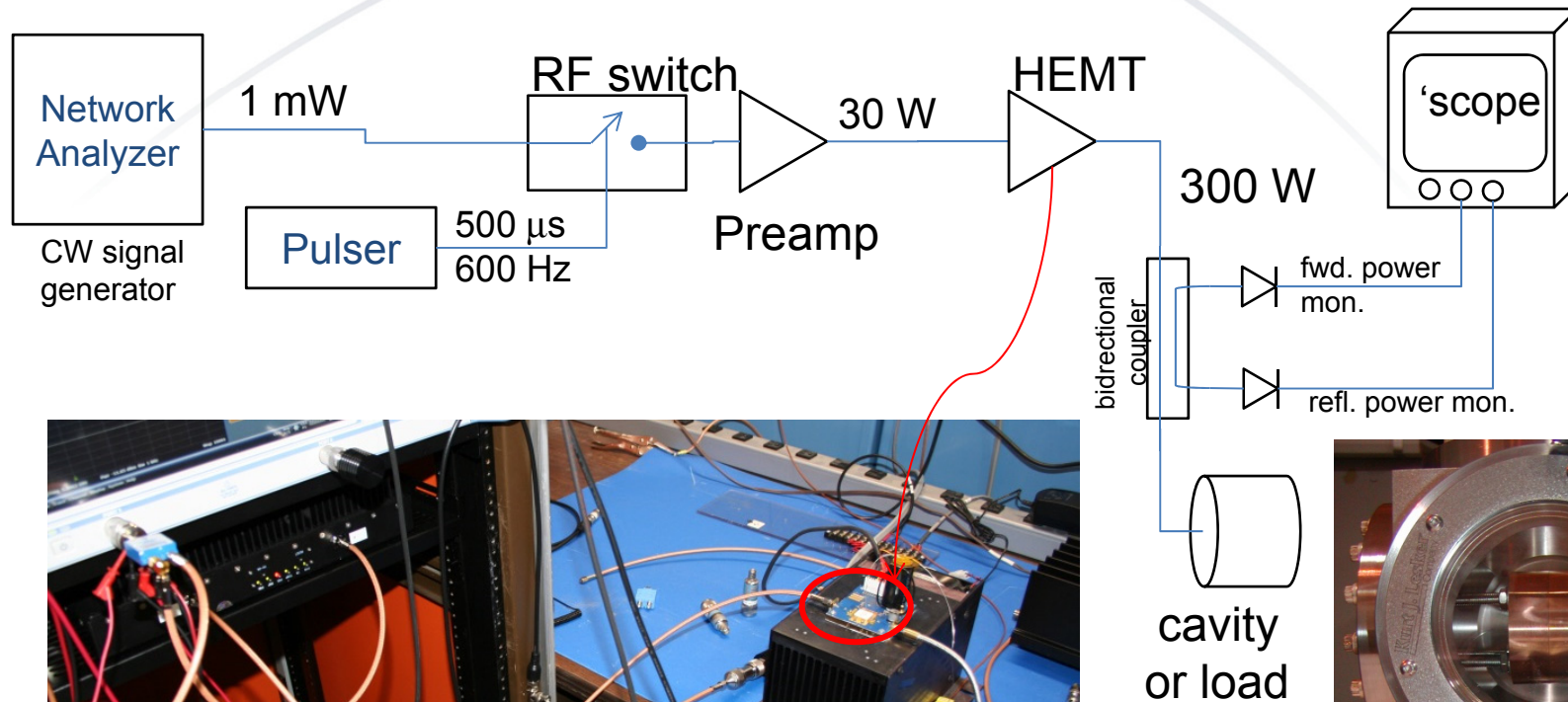
Autophase algorithm – adapts to different gradients, cell failure, cell spacing

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HEMT and Cavity Drive Testing



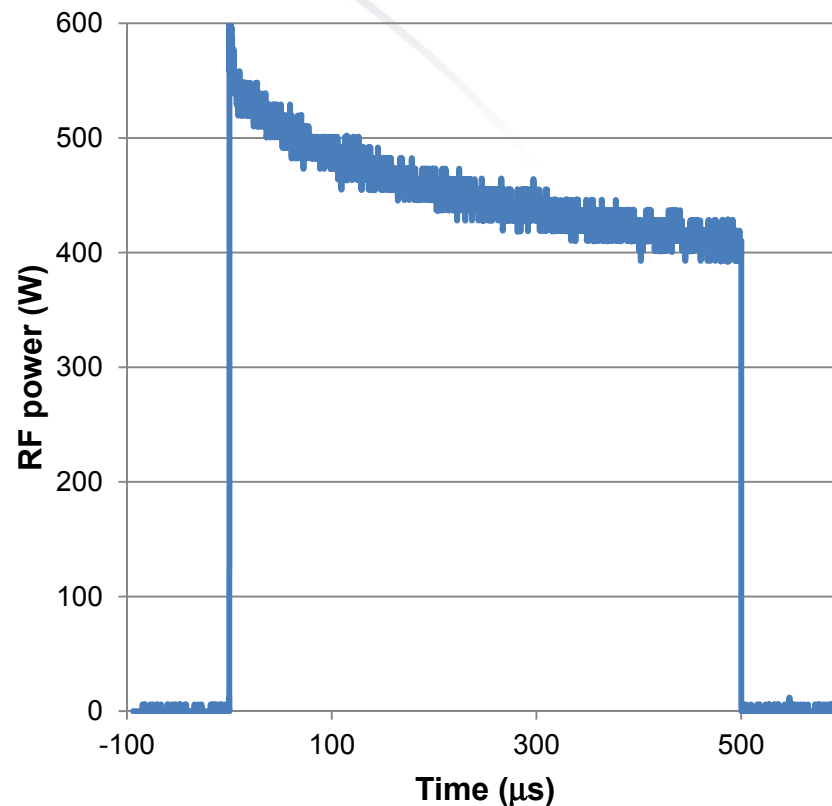
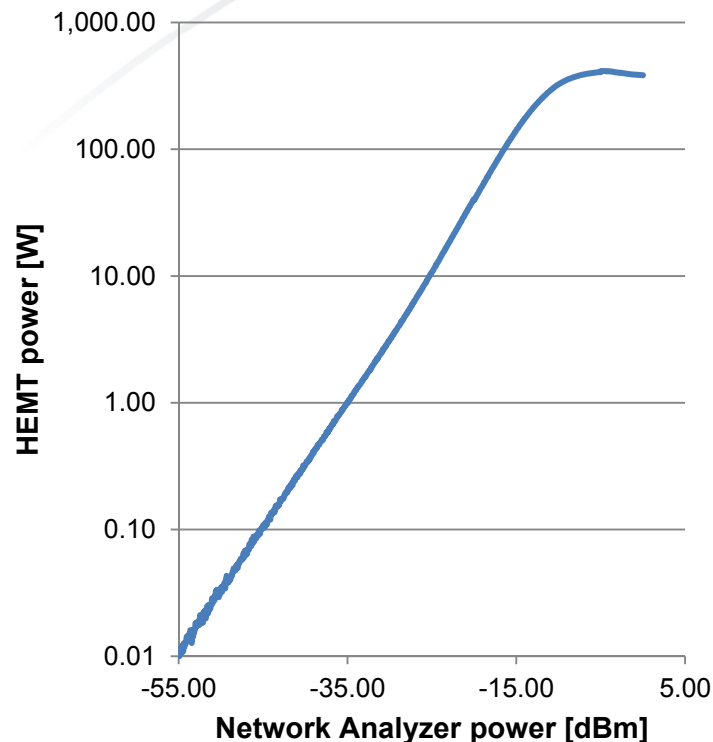
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HEMT Measurements

5.1 GHz



C-Band HEMT exceeds peak power specs at low DF
here: $(0.5 \text{ ms pulse length}) \cdot (10 \text{ ms period}) = 5\% \text{ DF}$

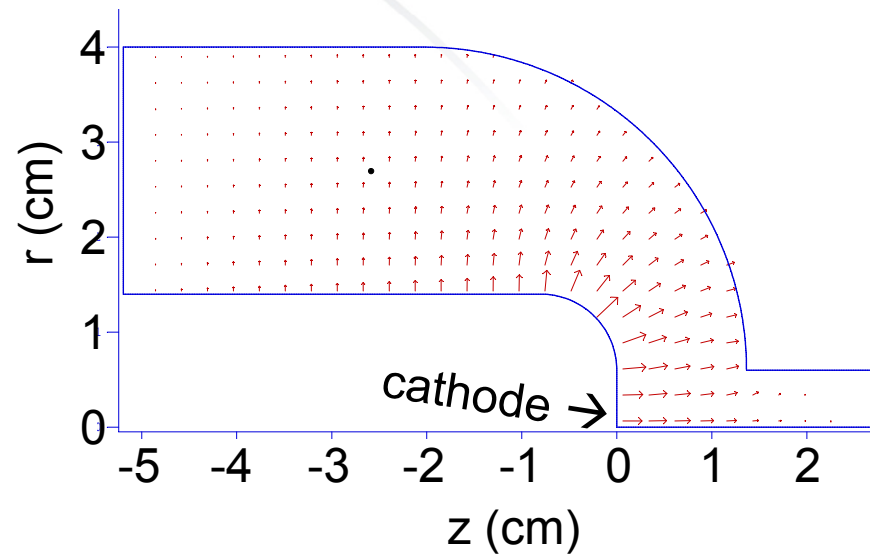
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Next Steps – Modeling & Sim.

- Beam source options
 - Subharmonic RF gun
 - Fundamental RF gun
 - e^- RFQ?
 - “Low-voltage” DC gun?
- More comprehensive beam dynamics
- Cavity coupler design



Conceptual “shorted-line” QW-like subharmonic electron gun

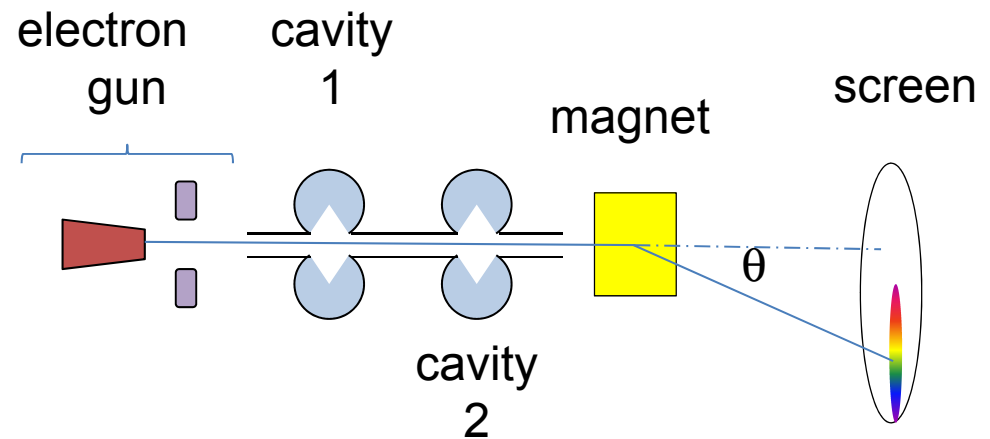
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Next Steps – Experiment

- Energy gain test
 - 20-kV DC gun
 - pulsed gun?
- Frequency tracking
- Autophasing
- Cavity coupler testing

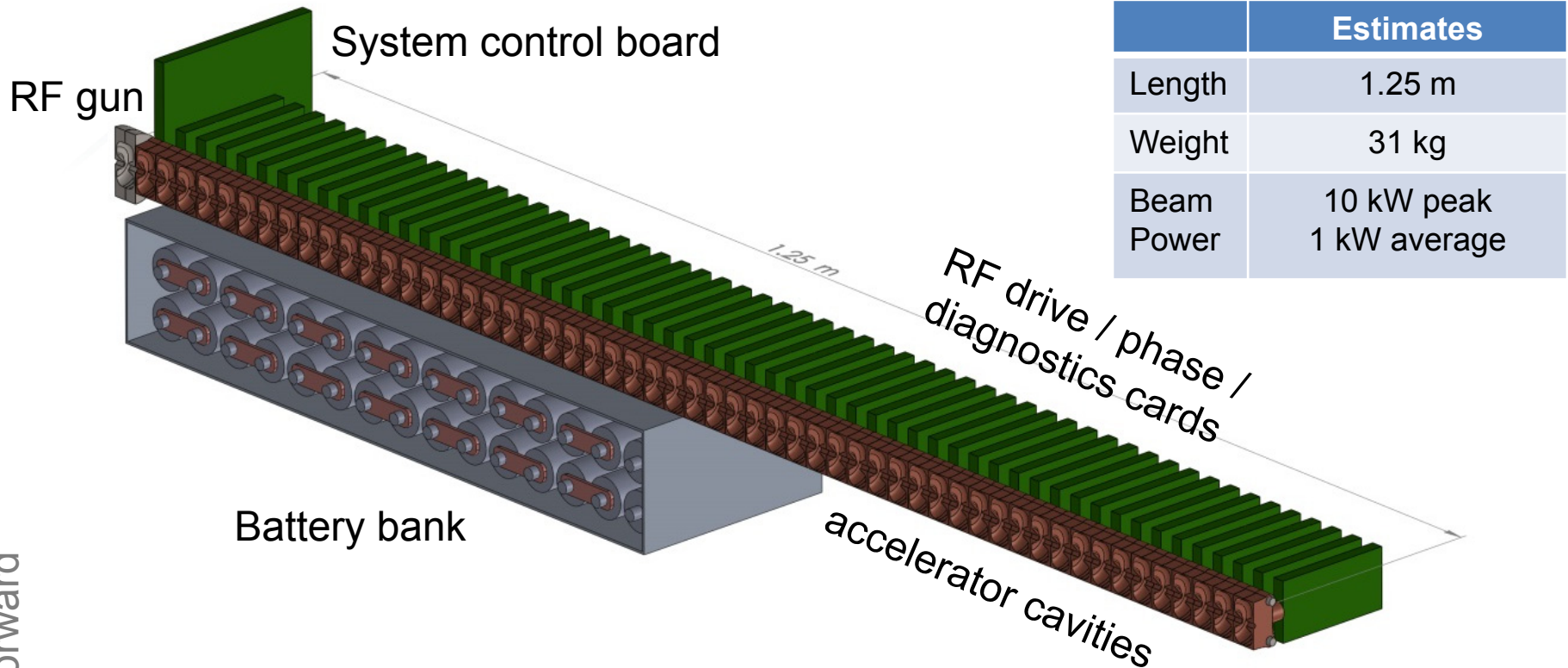


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Artist's Conception



Looking Forward

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Conclusions

- Accelerators in space can drive exciting new science, but are limited by achievable current & voltage
- RF-based linacs appear highly promising as alternatives for CONNEX
 - (relatively) efficient and compact
 - high V_{beam} for low V_{sat}
 - modular and highly redundant / fault tolerant
 - much work ahead, but no show-stoppers seen yet
- Terrestrial linacs may also be improved with the same technologies and approaches

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